



United International University

School of Science and Engineering

Final Examination; Year 2024; Trimester: Spring

Course: PHY 2105; Title: Physics; Sec: A-M

Full Marks: 40, Time: 2 Hours

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Questions no 1, 2, 3, 4 are mandatory to answer. Answer any one from question no 5 and 6.

1. (a) Initially $+q$ and $-q$ charges are separated at a distance r . If magnitude of the charges become doubled and distance between them increases r , what will be the change in attractive force? Justify your answer using necessary equations. 2 CO1
- (b) Two particles, one positively charged, and one negatively charged are separated at a distance r . From the concept of electric field lines, draw a diagram and discuss the possible locations on the axis where net electric field will be zero. 2 CO1
- (c) Which principal is used in microwave cooking? Show dipole oscillations in time varying electric field using necessary figures. 2 CO1

2. (a) In the figure 01, the four particles form a square of edge length $a=5.00$ cm and have charges $q_1 = +10.00$ nC, $q_2 = -20.0$ nC, $q_3 = +20.0$ nC, and $q_4 = -10.0$ nC.

Calculate the

- (i) magnitude of net electrostatic force on the charge q_4 at origin
- (ii) calculate the direction of the net electrostatic force on the charge q_4 at origin

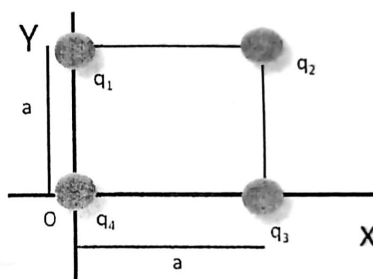


Fig. 01

- (b) Two charges $+6 \mu\text{C}$ and $+12 \mu\text{C}$ are placed 20 cm apart. Calculate the net electrostatic force on a test charge $+1 \mu\text{C}$ placed at the midpoint between the two charges. 2 CO3
- (c) How many charges will be available in 1 kg of electrons? (Consider mass of one electron, $m_e = 9.11 \times 10^{-31}$ kg and charge of one electron, $e = -1.6 \times 10^{-19}$ C) 2 CO3
3. (a) A charge of $-2.0 \mu\text{C}$ is located at the coordinates $y = 3$ cm while a second charge of $+2.0 \mu\text{C}$ is located at the coordinates $x = 3$ cm. Draw the charge arrangement and determine the magnitude and direction of the net electric field at the origin. 3 CO3
- (b) An electric dipole with the magnitude of dipole moment $P = 3 \times 10^{-8}$ Cm is found in an electric field with magnitude $E = 2000$ N/C. How much work must an external agent do to rotate this molecule by 180° in this field, starting from its fully aligned position, for which $\theta = 0^\circ$? 2 CO3
- (c) Two equal charges $q = 9 \times 10^{-9}$ C are placed at the two corners an equilateral triangle of side $r = 20$ cm. Draw the triangle with charges. Find the resultant electric field and its direction at the third corner of the triangle. 4 CO3
4. (a) Fig. 02 shows three charged particles with values $q_1 = +15 \mu\text{C}$, $q_2 = -87 \mu\text{C}$ and $q_3 = +66 \mu\text{C}$ and the values of distance x_1 , x_2 and x_3 are 5 cm, 18 cm and 27 cm from charge q_1 (at origin) respectively. Calculate the net electric potential at point P. 3 CO3

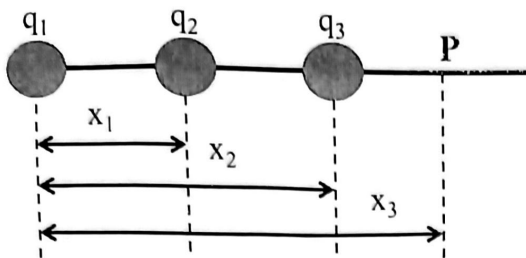


Fig. 02

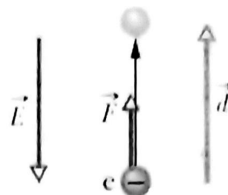


Fig. 03

(b) Electrons are continually being knocked out of air molecules in the atmosphere by cosmic-ray particles coming in from space in Fig. 03. Once released, each electron experiences an electric force due to the electric field that is produced in the atmosphere by charged particles already on Earth. Near Earth's surface the electric field has the magnitude $E = 600 \text{ N/C}$ and is directed downward.

(i) What is the change U in the electric potential energy of a released electron when the electric force causes it to move vertically upward through a distance $d = 510 \text{ m}$?

(ii) Through what potential change does the electron move?

(c) What is the electric potential at point P, located at the center of the square of charged particles shown in Fig. 4? Consider the distance d is 1.3 m , and the charges are,

$q_1 = +5\text{C}$, $q_2 = +10\text{C}$, $q_3 = +8\text{C}$, and $q_4 = +20\text{C}$.

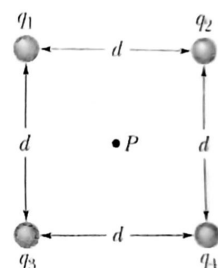


Fig. 04

5. (a) Three charged objects with charges $+5q$, $+10q$, and $+q$ are placed on x axis at origin, r , and $2r$ distance (place the charges from origin, respectively). If $+10q$ charged is now placed at $3r$ distance keeping the other charges in their positions, calculate the change in net electrostatic force on $+q$ charge.

(b) Show that, the magnitude of the electric field set up by an electric dipole at a distant point on the dipole axis (which runs through both particles) can be written in terms of either the product qd or the magnitude p of the dipole moment:

$$E = \frac{1}{2\pi\epsilon_0} \frac{p}{z^3}$$

where z is the distance between the point and the center of the dipole.

6. (a) Show that, at a distance r from an electric dipole with dipole moment magnitude $p = qd$, the electric potential of the dipole is

$$V = \frac{p \cos \theta}{4\pi\epsilon_0 r^2}$$

for $r \gg d$; the angle θ lies between the dipole moment vector and a line extending from the dipole midpoint to the point of measurement.

(b) Show that, potential for a line of charge with continuous charge distribution

$$V = \frac{\lambda}{4\pi\epsilon_0} \ln \left[\frac{L + \sqrt{L^2 + d^2}}{d} \right]$$

where the symbols have their usual meaning.

CO1: Define different physical quantities with examples. **CO2:** Derive/Show the various equations of electric field, electric potential, electric dipole, dipole moment, electrostatic force, etc. **CO3:** Evaluate different numerical problems based on the basic characteristics of electric charge, electric field, electric potential, electric dipole moment, and electrostatic force, etc.